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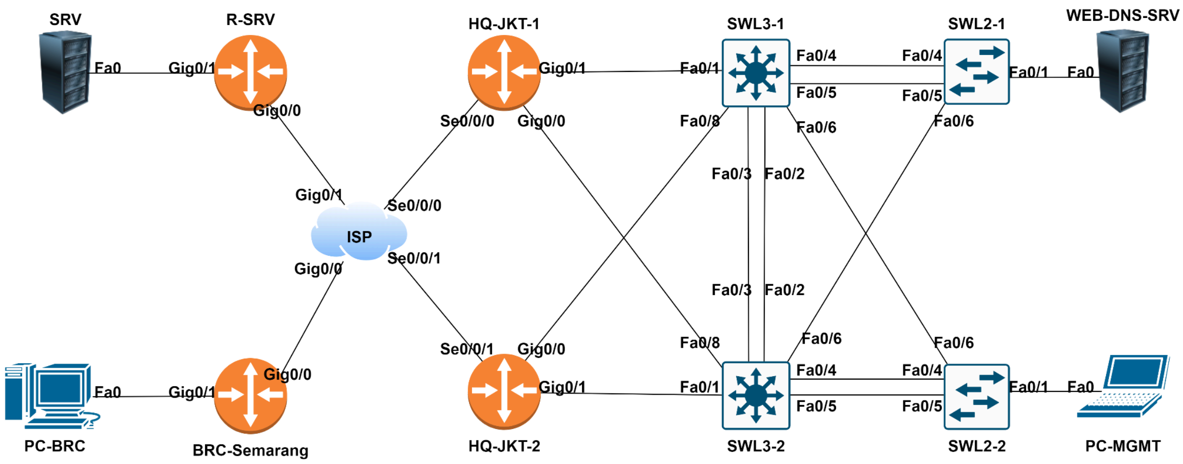
TEST PROJECT
MODULE A - NETWORK ENVIRONMENTS

IT NETWORK SYSTEM ADMINISTRATION

Network Topologi



A. Scenario

As a **Network Engineer**, you are asked to build an office network that has **two branches** with a Network Topology like the picture above with the aim that the **HQ-JKT** and **BRC-SEMARANG** offices can communicate with each other and access the server in the HQ-JKT office. For IP Address and Tasks follow the client's request below.

B. IP Address Documentation

DEVICE	INTERFACE	IP ADDRESS
ISP	GigabitEthernet0/0	103.17.222.1/30 2001:2025:AB:7D11::1/64
	GigabitEthernet0/1	201.50.11.1/30 2001:2025:AB:7D22::1/64
	Serial0/0/0	103.16.206.1/30 2001:2025:AB:7D33::1/64
	Serial0/0/1	103.99.207.1/30 2001:2025:AB:7D44::1/64
R-SRV	GigabitEthernet0/0	201.50.11.2/30 2001:2025:AB:7D22::2/64
	GigabitEthernet0/1	201.50.11.5/30
SRV	Fastethernet0	201.50.11.6/30
HQ-JKT-1	GigabitEthernet0/0	10.12.12.1/30
	GigabitEthernet0/1	10.11.11.1/30
	Serial0/0/0	103.16.206.2/30 2001:2025:AB:7D33::2/64
	Loopback0	172.16.0.1/32
	Tunnel0	10.192.168.1/30
HQ-JKT-2	GigabitEthernet0/0	10.12.12.5/30
	GigabitEthernet0/1	10.22.22.1/30

	Serial0/0/1	103.99.207.2/30 2001:2025:AB:7D44::2/64
	Loopback0	172.16.0.2/32
BRC-SMRG	GigabitEthernet0/0	103.17.222.2/30 2001:2025:AB:7D11::2/64
	GigabitEthernet0/1	10.1.10.1/24
	Loopback0	172.17.0.1/32
	Tunnel0	10.192.168.2/30
PC-BRC	Fastethernet0	10.1.10.100/24
SWL3-1	Fastethernet0/1	10.11.11.2/30
	Fastethernet0/8	10.12.12.6/30
	VLAN99	10.0.99.1/24
	VLAN100	10.0.100.1/24
	Loopback0	172.16.0.3/32
SWL3-2	Fastethernet0/1	10.22.22.2/30
	Fastethernet0/8	10.12.12.2/30
	VLAN99	10.0.99.2/24
	VLAN100	10.0.100.2/24
	Loopback0	172.16.0.4/32
SWL2-1	VLAN99	10.0.99.11/24
SWL2-2	VLAN99	10.0.99.12/24
PC-MGMT	Fastethernet0	DHCP
WEB-DNS-SRV	Fastethernet0	10.0.100.101/24

C. Tasks

You are assigned to do some tasks below correctly.

a. Basic Configuration

1. Configure device hostname according to the topology.
2. Configure domain name to "**jadi-juara2025.dki**".
3. Configure privileged mode password "**lks2025.dki**" on all devices.
4. Configure user "admin" with password "**lks2025.dki**" with maximum privileged.
5. Configure IPv4 and IPv6 address on all devices according to the table "**IP Address Documentation**".
6. Configure device timezone to GMT+7 on all devices.

b. Switching

1. Configure VTP version 2 on all switches. Configure SWL3-1 as the VTP Server. Other switches must as VTP Client. Configure VLAN on the VTP Server:
    a. VLAN 99 (MANAGEMENT)
    b. VLAN 100 (SERVER)
2. Configure link aggregation on switches refer to the list below. The L3 switches must act as a negotiator to perform link aggregation if using negotiation protocol, meanwhile the L2 switch must be configured as the responder.
    a. SWL3-1 <<=>> SWL3-2 (group-channel 1) without using any negotiation.
    b. SWL3-1 <<=>> SWL2-1 (group-channel 2) using open-standard protocol.
    c. SWL3-2 <<=>> SWL2-2 (group-channel 3) using Cisco proprietary protocol.
3. Configure ports are connected to the end device as the access port and enable portfast feature on the port.
4. Configure port connected to the switch as the trunk port and only allow the created VLAN to be passed.
5. Configure Spanning Tree Protocol with VLAN 100 to be root Primary and VLAN 99 to be root secondary in the device SWL3-1. meanwhile for SWL3-2 devices it can be configured in reverse.

c. Routing

1. Configure EIGRP routing on R-SRV and ISP.
    a. Use as number 2025.
    b. Advertise all installed network on R-SRV and ISP.
    c. Make sure both can reachable.
    d. Configure passive interface on an interface that does not require EIGRP updates.
2. Configure OSPF routing on HQ-JKT-1, HQ-JKT-2, SWL3-1, and SWL3-2.
    a. Use OSPF area 0 and instance number 2025.
    b. Router ID uses the IP of the loopback interface.
    c. Advertise all installed private networks including loopback interface and tunnel interface.
    d. Configure passive interface on specific interface that should not receive hello packets (interface leading to end user device).
    e. Ensure that Designated Router (DR) and Backup Designated Router (BDR) roles are not elected on a point-to-point OSPF link.
    f. Use command "default-information originate" to redistribute default route on HQ-JKT-1.
3. Configure default route on HQ-JKT-1 to ISP, HQ-JKT-2 to ISP, and BRC-Semarang to ISP.
4. Configure static routing on BRC-Semarang to reach the network IP of the WEB-DNS-SRV and PC-MGMT located in HQ-JKT, using the tunnel interface as the gateway. Do the same on HQ-JKT to reach the corresponding network in BRC-Semarang.

d. Services

1. Configure Network Address Translation for IPv4.
    a. Configure dynamic network address translation overload on HQ-JKT-1 and HQ-JKT-2 for the VLAN 99 and VLAN 100 addresses are translated into IPv4 address of interface connected to the ISP.
    b. Configure ACL to allow Network IP WEB-DNS-SRV and MANAGEMENT with use extended ACL name "PERMIT-NAT".
2. Configure FHRP on SWL3-1 and SWL3-2.
    a. Configure HSRP IPv4 version 2 on SWL3-1 and SWL3-2 for VLAN 99 and VLAN 100 with group number 2105.
3. Configure WEB and DNS Service on WEB-DNS-SRV.
    a. Enable WEB Service on WEB-DNS-SRV.

   b. Configure DNS Service on WEB-DNS-SRV with domain name "**lks.dki.net**".
4. Configure DHCP service for HQ-JKT.
   a. Configure DHCPv4 on HQ-JKT-1 for client VLAN 99.
   b. Use the virtual IP Router from FHRP as a gateway.
   c. Use IP Address from WEB-DNS-SRV as a DNS Server
   d. Enter IP Addresses are used by the device into the "Exclude-Address".
   e. Configure DHCP Relay on SWL3-1 and SWL3-2 for VLAN 99 with "ip helper address 172.16.0.1".
   f. Make sure PC-MANAGEMENT can reach WEB-DNS-SRV.
5. Configure NTP on all network devices.
   a. Configure NTP Server on NTP-SRV at SRV, and make sure HQ-JKT-1 and HQ-JKT-2 can reachable to NTP-SRV at SRV.
6. Configure Tunnell on HQ-JKT-1 and BRC-Semarang
   a. Configure GRE Tunnell to connect PC-BRC on BRC-Semarang with WEB-DNS-SRV on HQ-JKT.

e. Security

1. Configure port security on SWL2-1 and SWL2-2.
   a. Configure port security on SWL2-1 and SWL2-2 switch that is connected to hosts. Permit maximum 2 MAC Address and save those MAC Address on running-config. In case of policy violation, the port should not be err-disabled, but only save on syslog.
2. Configure SSH version 2 on BRC-Semarang, HQ-JKT, SWL3, and SWL2.
   a. Use local authentication.
   b. User with maximum privilege must be set to privileged mode after login.